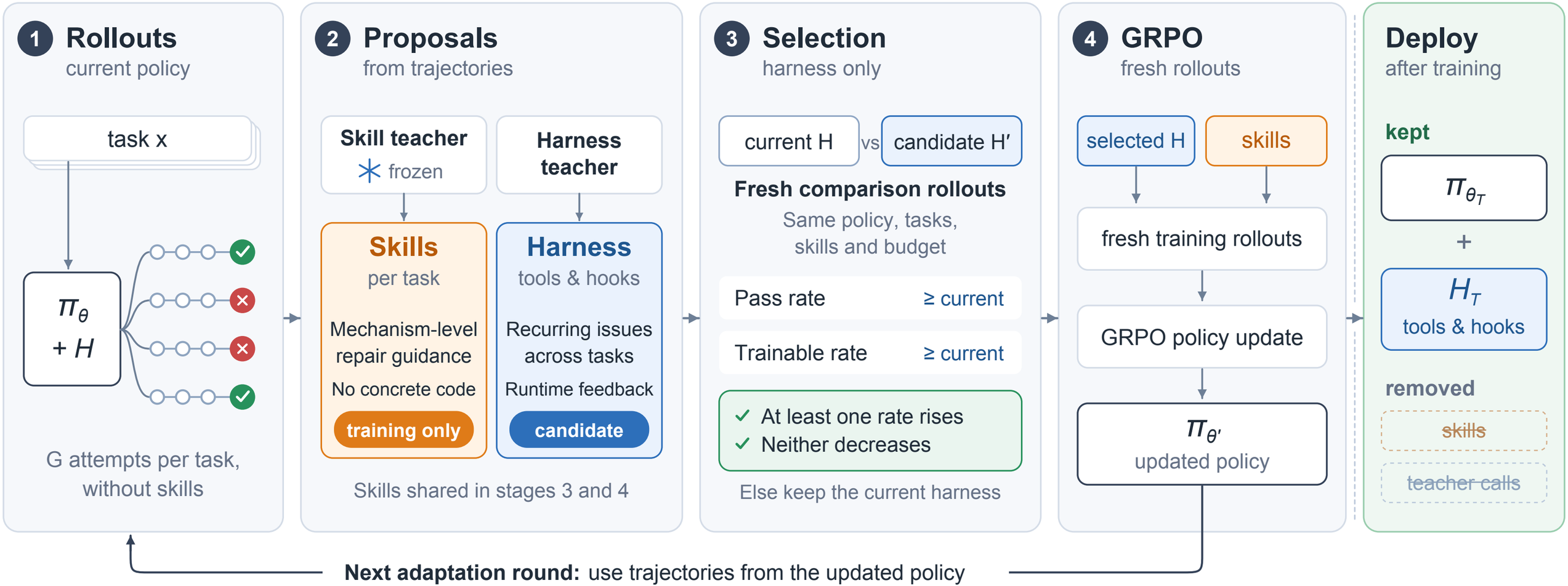

1 Rollouts
current policy
task x
π_θ + H
G attempts per task, without skills
2 Proposals
from trajectories
Skill teacher
frozen
Harness teacher
Skills
per task
Mechanism-level repair guidance
No concrete code
training only
Harness
tools & hooks
Recurring issues across tasks
Runtime feedback
candidate
Skills shared in stages 3 and 4
3 Selection
harness only
current H
vs
candidate H′
Fresh comparison rollouts
Same policy, tasks, skills and budget
Pass rate
≥ current
Trainable rate
≥ current
✓ At least one rate rises
✓ Neither decreases
Else keep the current harness
4 GRPO
fresh rollouts
selected H
skills
fresh training rollouts
GRPO policy update
$\pi_{\theta'}$
updated policy
Deploy
after training
kept
π_{θ_T}
+
H_T
tools & hooks
removed
skills
teacher calls
Next adaptation round: use trajectories from the updated policy